TECHNICAL PROJECT REPORT

Three-VSC Microgrid with Droop-Based Power Control

MATLAB/Simulink Control and Power-Sharing Model

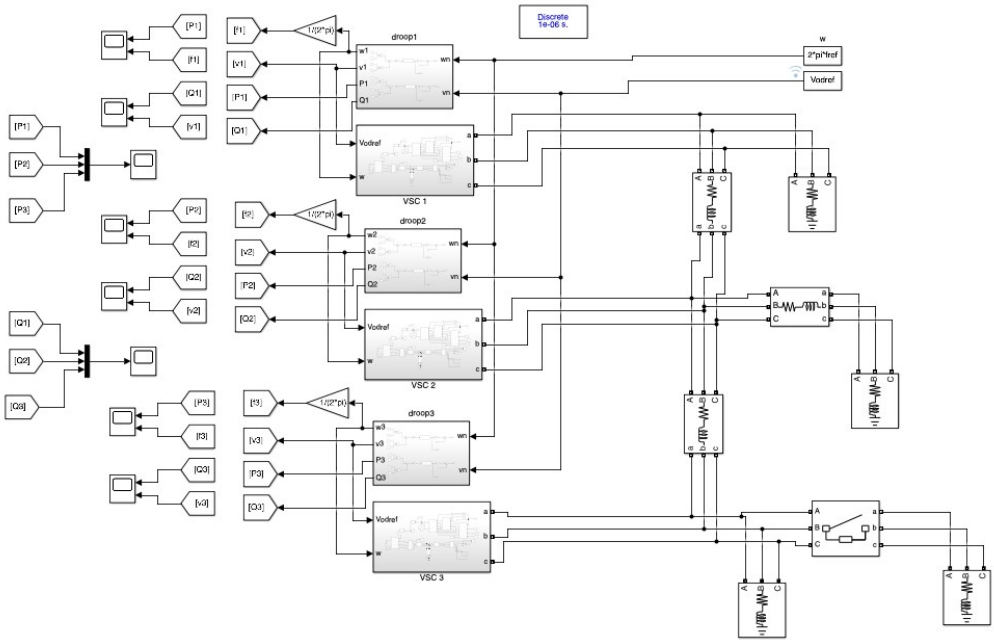


Figure 1. Top-level three-VSC microgrid model with decentralized droop controllers.

Prepared as a technical documentation of the simulation model

Abstract

This project presents a MATLAB/Simulink model of a three-voltage-source-converter (VSC) microgrid controlled using decentralized P–f and Q–V droop characteristics. Three parallel VSC units are connected to a common three-phase AC bus and are assigned independent droop controllers. Each converter uses an inner dq-frame current controller and an outer voltage controller, while measured three-phase voltages and currents are transformed into synchronous dq0 quantities for control. Active and reactive power are calculated in the dq frame and filtered before being fed to the droop equations. The model also includes a 700 V DC-link reference, 50 Hz grid-frequency reference, LCL/interface-filter parameters, and discrete simulation settings.

Project Scope

The report documents the model architecture, mathematical control strategy, converter control loops, dq transformation, droop implementation, initialization parameters, and the intended power-sharing behavior. The supplied model and subsystem screenshots are used as the primary technical evidence. Numerical transient-performance claims are intentionally not stated where corresponding waveform data were not supplied.

Keywords

Three-VSC microgrid; droop control; P–f control; Q–V control; grid-forming control; dq transformation; VSC; current controller; voltage controller; MATLAB/Simulink.

1. Introduction

Microgrids commonly combine multiple converter-interfaced sources and loads on a common AC bus. When several converters operate in parallel, the control system must regulate voltage and frequency while coordinating the sharing of active and reactive power. Communication-intensive centralized control can become complex as the number of converters increases. Droop control provides a decentralized mechanism where each converter modifies its frequency and voltage reference according to its measured active and reactive power.

In this project, three identical control structures are represented as VSC 1, VSC 2 and VSC 3. Each VSC has its own droop block and converter control subsystem. The common-bus arrangement allows the converters to participate in load sharing through local measurements.

2. System Architecture

The top-level model contains three parallel VSC branches connected to a common three-phase network. The principal signal flow is: three-phase measurements → dq0 transformation → power calculation and filtering → droop law → voltage/frequency references → outer voltage control → inner current control → modulation → VSC.

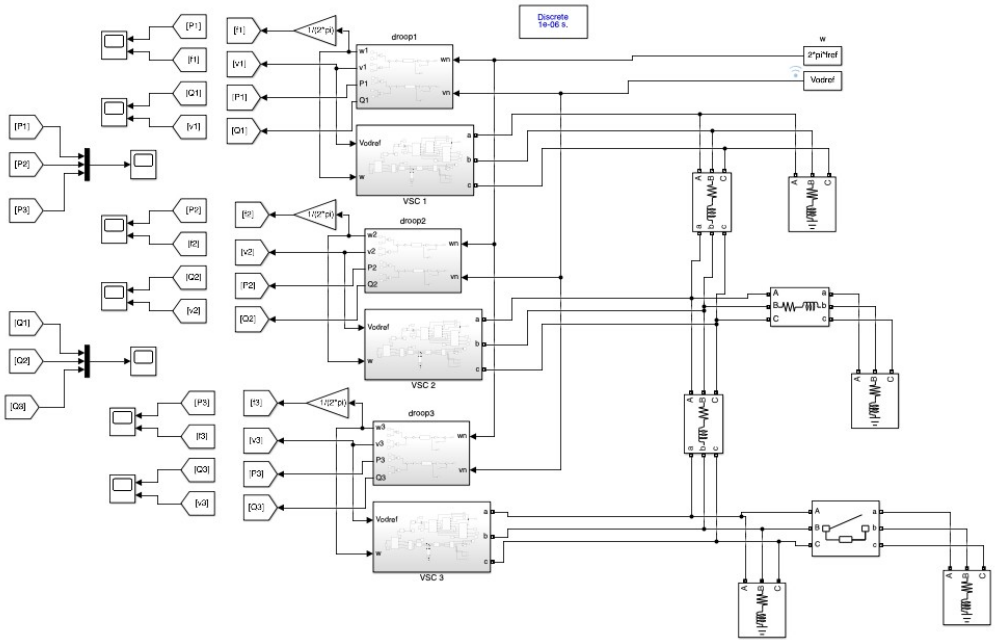


Figure 2. Complete top-level Simulink architecture.

2.1 Three Converter Units

VSC 1, VSC 2 and VSC 3 are connected in parallel to the same AC network. Each unit receives the common angular-frequency signal and voltage-reference information required by its control structure, while its droop controller produces converter-specific frequency and voltage references from locally calculated P and Q.

The model exposes P1/P2/P3, Q1/Q2/Q3 and corresponding voltage/current signals for observation. This arrangement is suitable for studying how converter power contributions change when the common load or operating condition changes.

2.2 Converter and Filter Interface

Each VSC is represented by a three-phase switching converter followed by an AC-side filter/interface. The supplied initialization parameters define the filter and converter-side quantities used by the control loops. The control structure is implemented in the synchronous rotating dq frame.

2.3 Model Initialization Parameters

Parameter	Value
Switching frequency, fsw	10 kHz
Fundamental/reference frequency, fref	50 Hz
Sampling/control interval, Ts	1/(fsw·fref)
d-axis voltage reference, Vodref	220√2 V
q-axis voltage reference, Voqref	0 V
DC-link reference, Vdc	700 V
Angular frequency, w0	2πfref
Filter inductance, Lf	3.5 mH
Filter resistance, Rf	0.1 Ω
Filter capacitance, Cf	50 μF
Interface inductance, Lc	0.35 mH
Interface resistance, Rc	0.05 Ω

Active-power droop coefficient, mp	5×10^{-5}
Reactive-power droop coefficient, np	3×10^{-3}
Power-filter reference parameter, fref/10	5 Hz

3. Droop-Based Power Control

The primary power-sharing mechanism is based on conventional frequency–active-power and voltage–reactive-power droop relationships. Instead of forcing every converter to maintain exactly the same internal reference under changing load, the converter reference is allowed to vary with its measured power.

3.1 Active-Power / Frequency Droop

The implemented concept can be represented by:

$$\omega_i = \omega_n - m_p P_i$$

where ω_i is the frequency reference generated for converter i, ω_n is the nominal angular frequency, P_i is the measured active power and m_p is the active-power droop coefficient. An increase in active-power output therefore produces a corresponding reduction in the converter's frequency reference.

3.2 Reactive-Power / Voltage Droop

The voltage/reactive-power relationship is represented by:

$$V_i = V_n - n_p Q_i$$

where V_i is the converter voltage reference, V_n is the nominal voltage reference, Q_i is the measured reactive power and n_p is the reactive-power droop coefficient. This permits the three converters to share reactive-power demand through their local voltage adjustments.

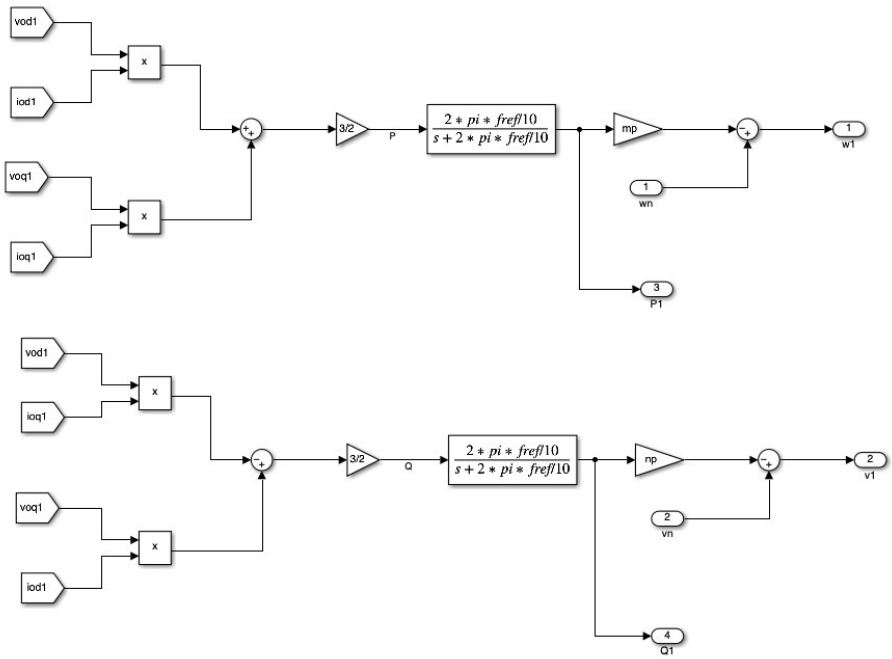


Figure 3. Active- and reactive-power calculation, filtering and droop-reference generation.

3.3 Power Measurement and Filtering

In the droop subsystem, d–q voltage and current components are multiplied to obtain the instantaneous active- and reactive-power terms. A 3/2 scaling factor is applied to the dq power expressions. The resulting P and Q signals are passed through first-order filters before entering the droop gains. Filtering prevents high-frequency switching components from directly driving the slow droop dynamics.

The power expressions used by the model follow the standard synchronous-frame form:

$$P = (3/2)(v_d\, i_d + v_q\, i_q)$$

$$Q = (3/2)(v_q\, i_d - v_d\, i_q)$$

4. Synchronous dq0 Transformation

The controller operates in a rotating reference frame because regulating DC-like d- and q-axis quantities is easier than directly regulating sinusoidal three-phase signals. The model converts measured phase quantities from abc coordinates to dq0 coordinates using an internally generated angular position.

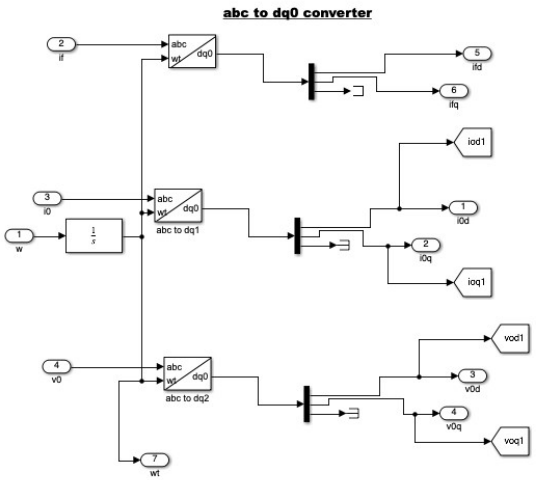


Figure 4. abc-to-dq0 transformation subsystem used for current and voltage measurements.

The model integrates the angular-frequency signal to obtain the electrical angle. Separate transformations are used for converter current, network/current-related quantities, and converter voltage. The resulting signals include id, iq, i0, vd, vq and v0. The d and q components are routed to the voltage and current controllers, while the transformed quantities are also used for power calculation.

5. VSC Control Structure

Each VSC uses a cascaded control structure consisting of an outer voltage loop and an inner current loop. The outer controller establishes the current references required to maintain the converter terminal voltage. The inner controller then tracks these current references and generates modulation commands for the switching converter.

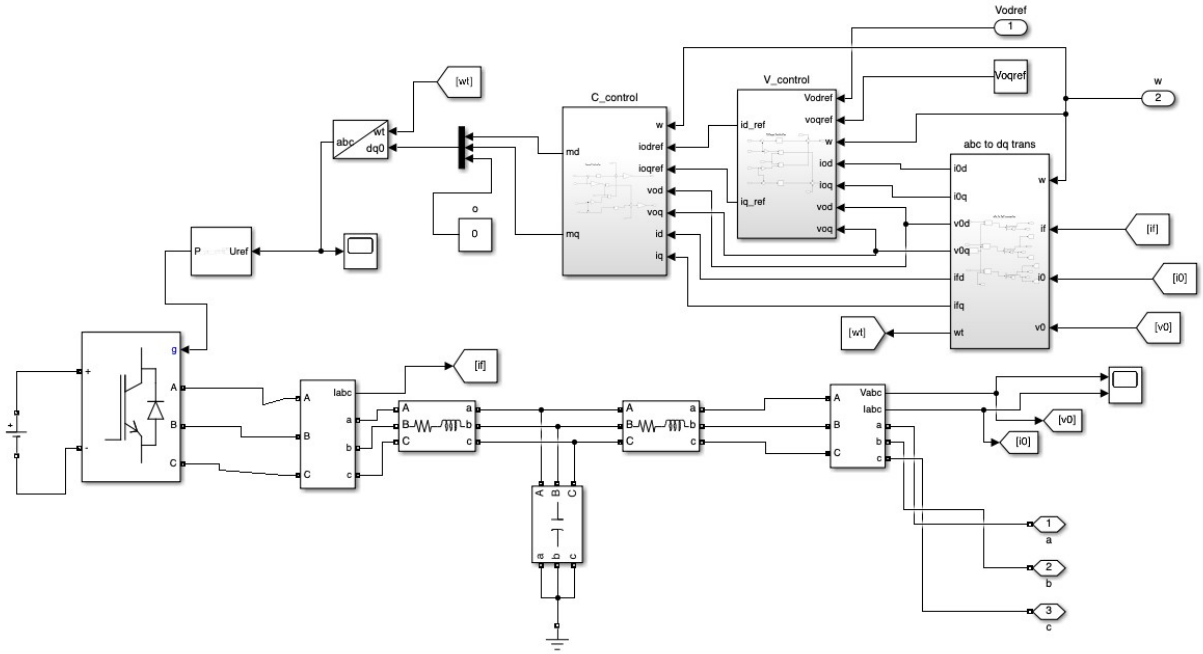


Figure 5. Internal VSC control structure showing voltage control, current control and dq transformation.

5.1 Outer Voltage Controller

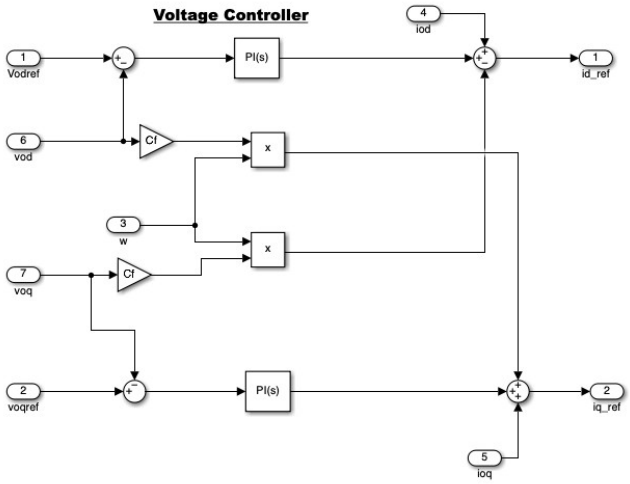


Figure 6. Voltage controller in the synchronous dq frame.

The voltage controller compares the d- and q-axis voltage references with measured voltage. PI regulators process the voltage errors and generate current references. The controller includes cross-coupling/feed-forward terms involving the filter capacitance, angular frequency and measured dq voltage components. The resulting id_ref and iq_ref signals are supplied to the inner current controller.

5.2 Inner Current Controller

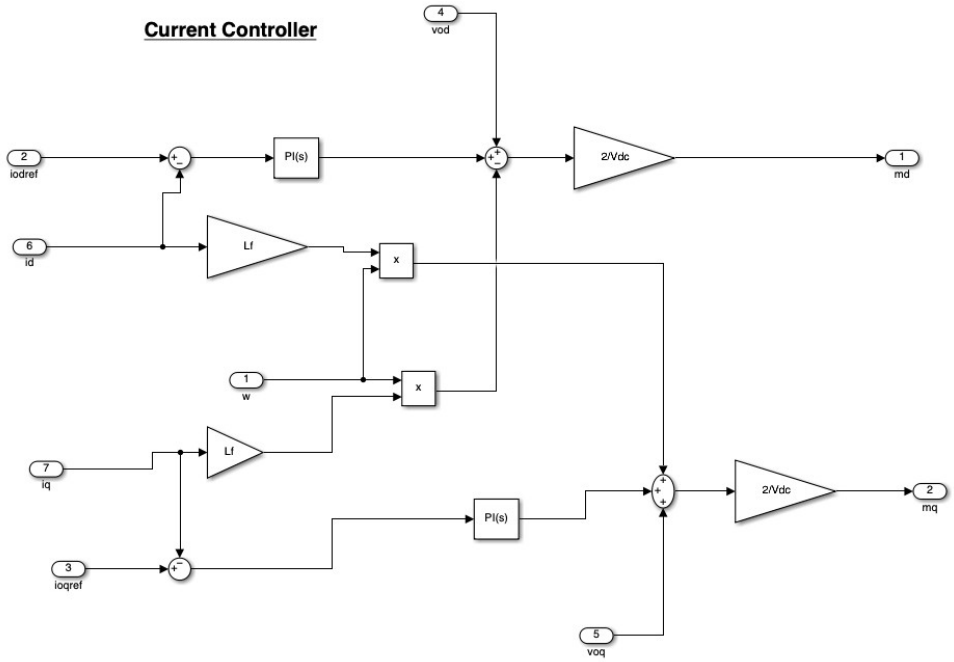


Figure 7. Inner dq current controller with decoupling terms and modulation scaling.

The current controller compares id_ref/iq_ref with measured id/iq and uses PI regulators to reduce the current tracking errors. The controller also contains cross-coupling compensation associated with the filter inductance and synchronous angular frequency. The resulting voltage commands are combined with the measured voltage terms and normalized by the DC-link voltage through the 2/Vdc scaling blocks to obtain modulation commands.

5.3 Control-Gain Initialization

The model initialization computes the controller gains from the filter parameters and selected control bandwidths. For the current loop, the script defines Tci = 4/(2πfsw), wci = 1/Tci, followed by Kpc = Lf·wci and Kic = Rf·wci. The voltage-loop quantities are generated from the selected phase margin and derived crossover frequencies. This makes the controller parameters traceable to the model's physical and control settings.

6. Complete Control Signal Flow

The complete sequence for each converter can be summarized as follows:

1. Measure three-phase converter/network voltage and current.

2. Transform abc quantities to synchronous dq0 coordinates using the electrical angle.

3. Calculate active and reactive power from dq quantities and apply first-order filtering.

4. Apply P–f and Q–V droop laws to generate converter-specific frequency and voltage references.

5. Use the outer voltage PI controllers to generate id_ref and iq_ref.

6. Use the inner current PI controllers and dq decoupling terms to generate modulation commands.

7. Feed the modulation signals to the three-phase VSC switching stage.

8. Connect all three VSC outputs to the common AC network so that power is shared according to the droop characteristics.

7. Simulation Configuration

The supplied model contains a discrete simulation configuration and a switching/control frequency of 10 kHz. The initialization script sets the fundamental reference frequency to 50 Hz and the DC-link reference to 700 V. The controller is therefore designed around a 50 Hz three-phase system with converter-side switching dynamics.

Because the uploaded material contains the model architecture and subsystem diagrams but does not include a complete set of final waveform screenshots, this report treats waveform performance as a validation step rather than assigning unverified numerical settling times, overshoot values, THD values or power-sharing errors.

8. Expected Operating Behavior

Under steady operating conditions, the three VSCs should establish compatible voltage and frequency references and contribute to the common load. If active-power demand changes, the P–f droop loops should redistribute active power among the converters according to their droop characteristics. A reactive-power change should similarly produce voltage-reference adjustments through the Q–V droop loops.

The inner current loop is expected to be faster than the outer voltage and droop dynamics. This time-scale separation allows the converter to track current commands while the slower supervisory loops determine the operating point.

9. Validation and Recommended Measurements

For a complete numerical validation, the following signals should be recorded from the supplied model:

- Three-phase bus voltage and current waveforms.
- VSC-1, VSC-2 and VSC-3 active powers P1, P2 and P3.
- VSC-1, VSC-2 and VSC-3 reactive powers Q1, Q2 and Q3.
- Frequency/angular-frequency references w1, w2 and w3.
- Voltage references v1, v2 and v3.
- d-axis and q-axis voltage/current components.
- DC-link voltage and converter modulation signals.
- Power-sharing response before and after a load step or breaker operation.

A useful validation sequence is to start all three converters under the same nominal references, introduce a controlled load change, and compare P1–P3 and Q1–Q3. The key checks are stable voltage/frequency, bounded current, reasonable transient response, and consistent power redistribution among the three VSCs.

10. Model Limitations and Assumptions

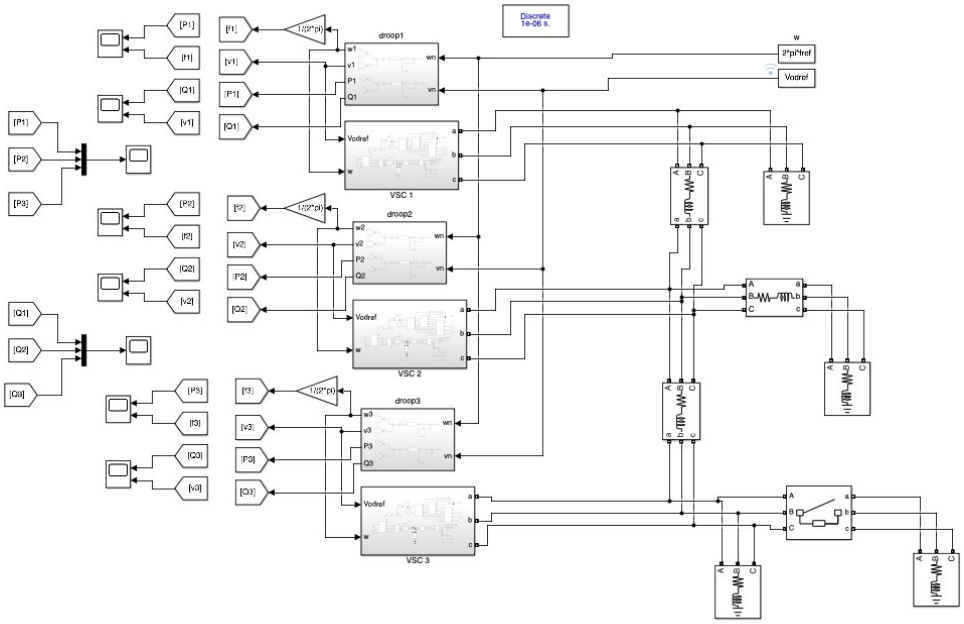
- The report documents the supplied simulation model and does not claim hardware validation.
- The switching devices are represented at the Simulink model level; semiconductor loss and thermal behavior are not the focus.
- Exact load-step values and final waveform measurements were not provided with the screenshots, so numerical performance metrics are not asserted.
- Secondary restoration of voltage/frequency and communication-based tertiary control are outside the demonstrated primary droop-control scope.
- The droop coefficients are model parameters and should be tuned with respect to converter ratings and desired power-sharing characteristics in a practical implementation.

11. Conclusion

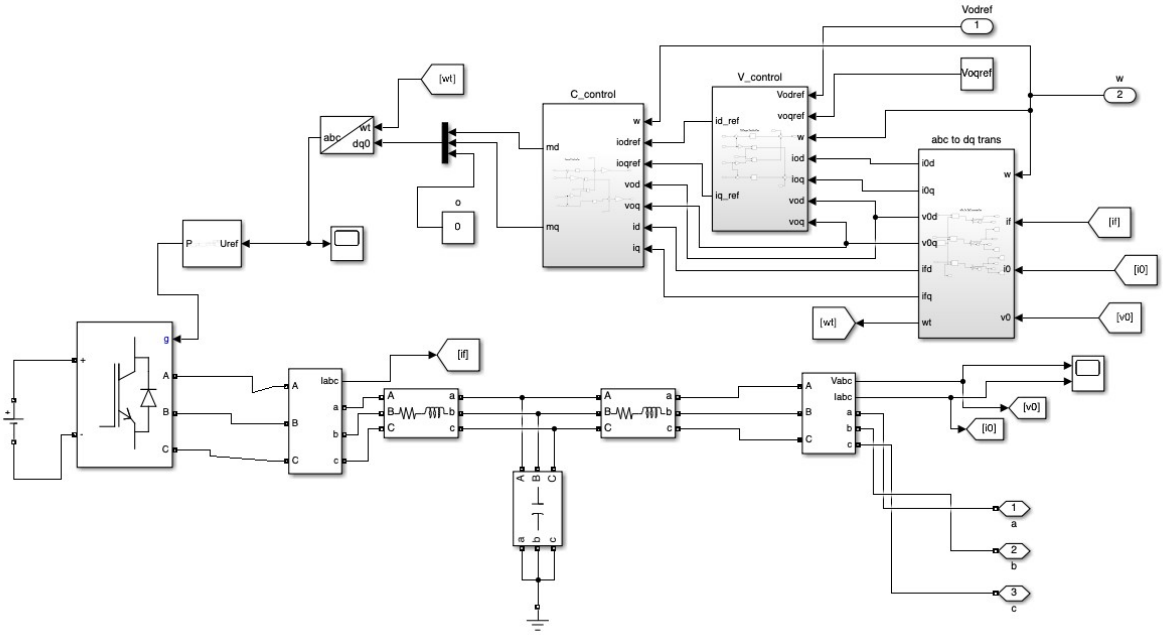
The developed MATLAB/Simulink model demonstrates a three-VSC microgrid using decentralized droop-based power control. The architecture combines P–f and Q–V droop characteristics with dq-frame voltage and current regulation. The cascaded voltage/current controller structure, dq0 transformations, filtered power calculation and three parallel VSC branches form a coherent framework for studying inverter-based microgrid operation and power sharing.

The model is suitable as a foundation for further work such as load-step testing, droop-coefficient tuning, secondary frequency/voltage restoration, unequal converter ratings, harmonic analysis, and hardware-oriented controller implementation.

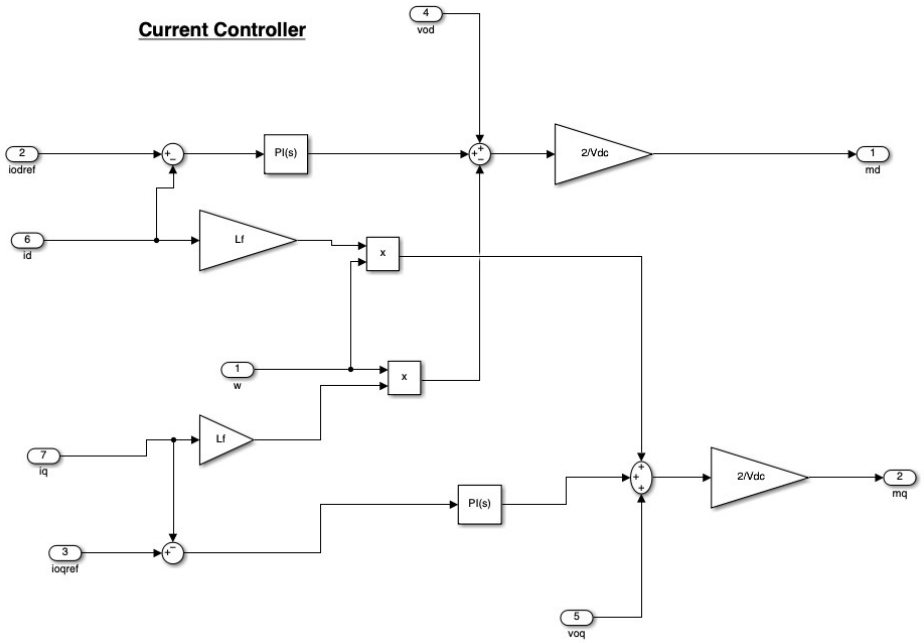
Appendix A – Supplied Model Figures



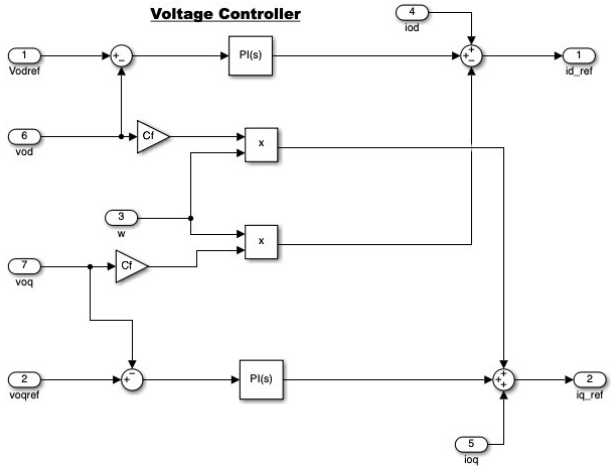
01 Microgrid Top Level



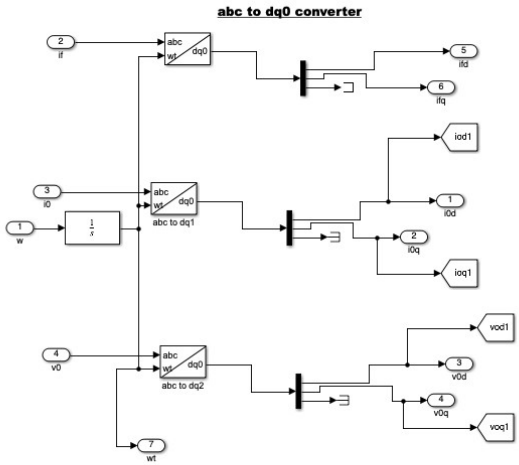
02 Vsc Control Structure



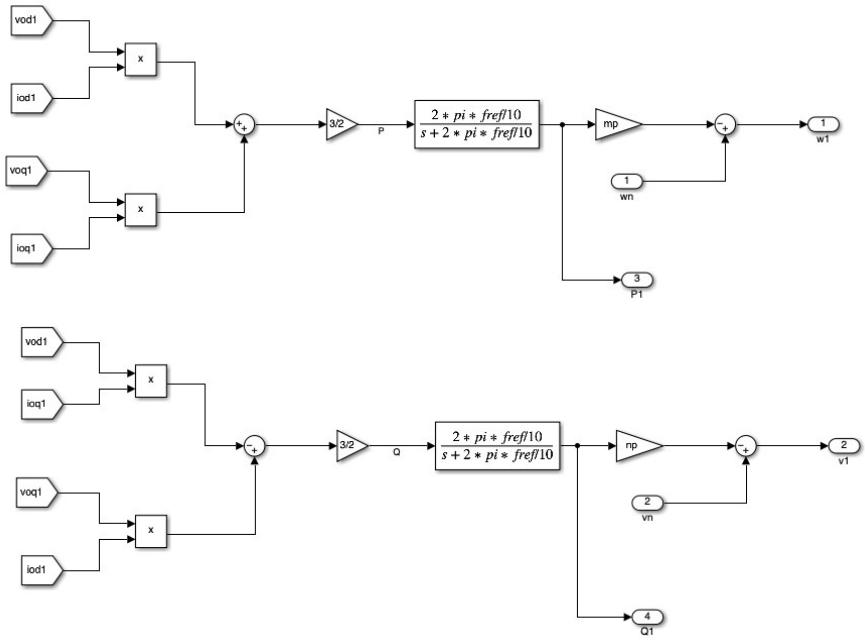
03 Current Controller



04 Voltage Controller



05 Abc Dq0 Transformation



06 Droop Controller